



Probability II

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 Stat 141 | Fall 2026
 Wednesday, Week 3

Announcements/Reminders

- Quiz tomorrow during lab
 - Arrive on time to use the full 30 mins
 - Bring something to write with
 - Be mindful of others when arriving / if finished early
- Math/Stats colloquium tomorrow - student summer research!

Last Lecture

- Introduced **probability theory**
- Saw the **Law of Large Numbers**
- Defined and practiced using key **rules of probability**
- Introduced **conditional probability**

Goals for Today

- Review **Conditional Probability**
- Define **Independence** and the **Multiplication Rule**
- Practice using the **Law of Total Probability** and **Bayes' Rule**

Conditional Probability

Conditional Probability: probability of something occurring, **given** that another event has **already** occurred.

We write the *conditional probability of Event A given Event B has occurred* as,

$$P(A | B)$$

Example from the end of last class,

- Question: What is the probability that a random *sophomore* preferred *coffee*?
- Event A: The student prefers coffee
- Event B: The student is a sophomore
- Answer: $P(\text{Coffee} | \text{Sophomore}) = P(A|B)$

Conditional Probability

How do we calculate conditional probabilities?

In general,

Theorem: Conditional Probability Rule

The Conditional Probability of an event A given another event B is

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

$$P(\text{Coffee}|\text{Sophomore}) = \frac{P(\text{Sophomore and prefers Coffee})}{P(\text{Sophomore})}$$

- The probability that a student who is a sophomore (B) prefers coffee (A):

$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{25}{100} \bigg/ \frac{45}{100} = \frac{25}{45}$$

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Activity from last class : Coffee or Tea?

	Coffee	Tea	total
First-year	7	10	17
Sophomore	25	20	45
Junior	13	12	25
Senior	8	5	13
total	53	47	100

- Event A : The student prefers coffee
- Event B : The student is a sophomore

1. **Probability student prefers coffee:** $P(\text{Coffee}) = P(A) = \frac{53}{100}$

2. **Probability student is a sophomore:** $P(\text{Sophomore}) = P(B) = \frac{45}{100}$

3. **Probability student is a sophomore and prefers coffee:** $P(\text{Sophomore and Coffee}) = P(A \text{ and } B) = \frac{25}{100}$

4. **Probability a random sophomore prefers coffee:**

$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{25}{100} \bigg/ \frac{45}{100} = \frac{25}{45}$$

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Using Conditional Probability

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Using Conditional Probabilities...

In the next few slides, we're going to use the concept of conditional probability to explore 4 more concepts:

1. Independence
2. Multiplication Rule
3. Law of Total Probability
4. Bayes' Rule

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Independence

We say two events are **independent** if knowing one occurs doesn't change the probability that another occurs.

- For example, rolling a 3 on a die (*event B*) doesn't change the probability that it rains today (*event A*), so these two events are independent.
- Learning that an even number has been rolled on a die (*event B*) does change the probability that a 4 was rolled (*event A*), so these two events are not independent.

Theorem: Criteria for Independence

Two events A and B are independent if and only if

$$P(A|B) = P(A) \quad \text{and} \quad P(B|A) = P(B)$$

Q: Were the events “preferring coffee” (A) and “being a sophomore at Reed” (B) independent?
No! That's because $P(A) = \frac{53}{100}$ and $P(A|B) = \frac{25}{45}$, so $P(A) \neq P(A|B)$

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Multiplication Rule

What if we want to know the probability that two events **both** occur?

One approach: Rearrange our formula for conditional probability!

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$
$$\Rightarrow P(A|B)P(B) = P(A \text{ and } B)$$

Theorem: Multiplication Rule

For any events A and B ,

$$P(A \text{ and } B) = P(A|B) \times P(B)$$

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Multiplication Rule

Theorem: Multiplication Rule

For any events A and B ,

$$P(A \text{ and } B) = P(A|B) \times P(B)$$

In the previous example...

- Event A : The student prefers coffee
- Event B : The student is a sophomore

and

- $P(B) = \frac{45}{100}$
- $P(A|B) = \frac{25}{45}$
- $P(A \text{ and } B) = \frac{25}{100}$

Does it work?

$$P(A|B)P(B) = \left(\frac{25}{45}\right) \left(\frac{45}{100}\right) = \frac{25}{100}$$

Checks out!

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Multiplication Rule - Independent Events

Theorem: Multiplication Rule

For any events A and B ,

$$P(A \text{ and } B) = P(A|B) \times P(B)$$

Q: If events A and B are independent, what happens to the multiplication rule?

Theorem: Multiplication Rule for Independent Events

For any events A and B that are **independent**,

$$P(A \text{ and } B) = P(A) \times P(B)$$

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The Law of Total Probability

Consider two events:

- Event A : The student prefers coffee
- Event B : The student is a sophomore

In this scenario, notice for any student who prefers coffee (A), it's possible...

- they're a sophomore (B)
- they're **not** a sophomore (B^c)

Thus,

$$\begin{aligned} P(A) &= P(A \text{ and } B) + P(A \text{ and } B^c) && \text{(Addition Rule)} \\ &= P(A|B)P(B) + P(A|B^c)P(B^c) && \text{(Multiplication Rule)} \end{aligned}$$

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The Law of Total Probability

Theorem: The Law of Total Probability

Let A and B be events. Then

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

- The Law of Total Probability is useful for finding “overall” probabilities when we only have the “pieces”.
- We'll see a real-world example next!

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Patient Health

Suppose you're a medical researcher interested in population health.

- 80% of individuals see a physician each year
- 20% do not
- Among people who see a physician, 95% are generally healthy.
- Among people who *don't* see a physician, 70% are generally healthy.

Q: What proportion of adults are generally healthy?

Let

- Event A : Individual is healthy
- Event B : Individual sees a physician each year
- Event B^c : Individual doesn't see a physician each year

This question is asking: $P(A) = ???$

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Patient Health

Theorem: The Law of Total Probability

Let A and B be events. Then

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

We want to answer $P(\text{Healthy})$.

Using the Law of Total Probability:

$$\begin{aligned} P(\text{Healthy}) &= P(\text{Healthy}|\text{Doctor})P(\text{Doctor}) + P(\text{Healthy}|\text{Doctor}^c)P(\text{Doctor}^c) \\ &= P(A|B)P(B) + P(A|B^c)P(B^c) \\ &= (.95)(.80) + (.70)(.20) \\ &= 0.90 \end{aligned}$$

Thus, 90% of adults are generally healthy!

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Bayes' Rule

What if we know $P(B|A)$, but we want to know $P(A|B)$?

Theorem: Bayes' Rule

Let A and B be events. Then,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Proof: From the definition of multiplication rule,

$$P(A \text{ and } B) = P(A|B) \times P(B)$$

$$P(A \text{ and } B) = P(B|A) \times P(A)$$

Setting these two equations equal, we find

$$\begin{aligned} P(A|B) \times P(B) &= P(B|A) \times P(A) \\ \Rightarrow P(A|B) &= \frac{P(B|A)P(A)}{P(B)} \end{aligned}$$

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Activity

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Activity 1: Testing for a Rare Disease

Suppose that we have a rapid COVID-19 test where:

- Given a person **does not** have COVID, the test is (correctly) negative 99% of the time.
- Given a person **does** have COVID, the test is (correctly) positive 80% of the time.

Assume that the overall prevalence of COVID at the time of the test was 1%.

Q: Suppose a person takes this test and receives a positive diagnosis – **what is the probability that the person has COVID?**

- **Hint 1:** Write out the object in question in terms of a conditional probability
- **Hint 2:** Define the relevant events and their complements in this setting
- **Hint 3:** Write out all the info given in terms of probabilities and events A and B .
- **Hint 4:** Start with Bayes' Rule! You'll also need the Law of Total Probability!

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Bonus Activity: Bayes' Rule

Theorem: Bayes' Rule

Let A and B be events. Then

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Q: Consider Bayes' Rule. Under what circumstances will $P(A|B) = P(B|A)$?

Q: Suppose $P(B|A) = 1$:

- What does this suggest about A and B ?
- What is $P(A|B)$ in this case?

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Story time: Independence and “Lumpy” Randomness

A professor from the Georgia Institute of Technology asked students on their homework assignment to either flip a coin 200 times and record the results, or pretend to flip a coin and fake 200 results. After HW submission, students were amazed that he could correctly identify whether everyone’s results were real or fake...

- Was he omniscient?
- Nope! His strategy was simply to look for runs of at least six heads or six tails in a row. **Why did this work?**
 - Independence and a large number of trials means a run of six is very likely from real coin flips.
 - Humans writing pretend coin flips are influenced by previously written outcomes.
- This idea has connections to many topics, ranging from video games to fraud detection!

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Recap

Theorem: Criteria for Independence

Two events A and B are independent if and only if

$$P(A|B) = P(A) \quad \text{and} \quad P(B|A) = P(B)$$

In words: A and B are independent if and only if knowing about B tells us nothing about the probability of A , and vice versa

Theorem: Multiplication Rule

For any events A and B ,

$$P(A \text{ and } B) = P(A|B) \times P(B) = P(B) \times P(A|B)$$

Can think: A and B means that first B happens, then **given** that B happens, A also happens.

Theorem: The Law of Total Probability

Let A and B be events. Then

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

Think about breaking A into two disjoint pieces: (A and B) or (A and B^c) [addition rule]. Then we apply the multiplication rule to each piece.

Theorem: Bayes' Rule

Let A and B be events. Then,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

This follows algebraically from conditional probability and the multiplication rule.

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